

LET'S LEARN PYTHON

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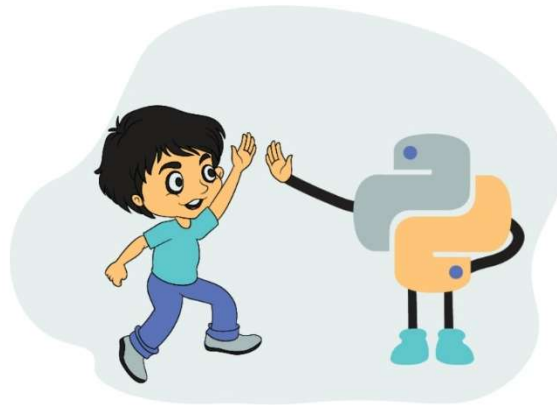
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Introduction

What is Python?

Created by Guido van Rossum, Python is an object-oriented, general-purpose, high-level programming language.

The main aim of Python was to make programming easier for beginners and allow developers to get things done in fewer lines of code compared to other programming languages. Because of this simplicity, Python has become one of the most popular choices for learning programming in recent times.



Python is:

- General-purpose → can be used in web development, data science, AI, automation, and more.
- Interpreted → code runs line by line, no need for compilation.
- Interactive → you can test code directly in its shell (REPL).
- Object-Oriented → supports classes and objects for structured programming.
- Readable → uses English keywords and simple syntax rather than complex symbols.

Python is a perfect choice of programming language if you are a beginner. If you have any previous programming experience, then learning Python would be a cakewalk.

History of Python:

- 1985–1990: Guido van Rossum started working on Python at CWI (Centrum Wiskunde & Informatica) in the Netherlands.
- Inspiration: Python was designed as a successor to the ABC programming language. ABC had good features like exception handling and worked with the Amoeba operating system, but it also had some limitations.



- Guido wanted to keep the good parts of ABC while fixing its flaws. This led him to create a new scripting language that was simple, powerful, and practical.
- 1991: Python was officially released. The first version already included important features such as functions, exception handling, and core data types like strings, lists, and dictionaries.
- The Name “Python”:

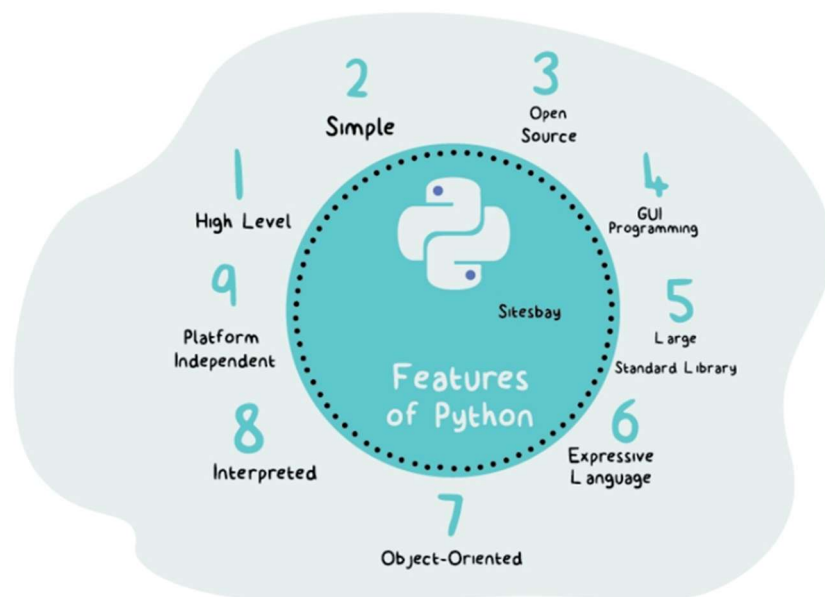
The name did not come from the snake 🐍. Guido was a big fan of the BBC comedy show “Monty Python’s Flying Circus”. He wanted a name that was short, unique, and slightly mysterious, so he chose Python.



And that’s how one of the world’s most loved programming languages was born. Amazing, right?

Features of Python:

- Easy-to-learn - Python has few keywords, simple structure, and a clearly defined syntax. This makes it easy to catch for beginners.
- Easy-to-read - Python code is more clearly defined and visible to the eyes.
- Easy-to-maintain - Python's source code is fairly easy-to-maintain
- Broad standard library - Python's bulk libraries are very portable and cross-platform compatible with UNIX, Windows, and Macintosh.
- Interactive Mode - Python has support for an interactive mode that allows interactive testing and debugging of snippets of code.
- Portable - Python can run on a wide variety of hardware platforms and has the same interface on all of those.
- Extendable - You can add low-level modules to the Python interpreter. These modules enable programmers to add or customize their tools to be more efficient



Why Learn Python?

Do you remember that Python is a general-purpose programming language?

That means it can be used almost everywhere — and this is exactly why developers around the world love Python so much.

Python is used for:

- Web Development
- App Development
- Data Science
- Machine Learning
- Internet of Things (IoT)

...and the list goes on!

Let's deep-dive into some of these areas:

Web Development

Python is one of the top choices for backend web development (not frontend).

Django → A powerful framework used for building large, full-fledged applications.

Flask → A lightweight framework suitable for smaller applications.

Both frameworks are in high demand, and developers skilled in Django or Flask often get high-paying roles.



App Development

Python can also be used for mobile applications.

While not the most popular choice for mobile apps,

Python has a framework called Kivy, which allows you to build cross-platform apps (apps that run on both Android and iOS).

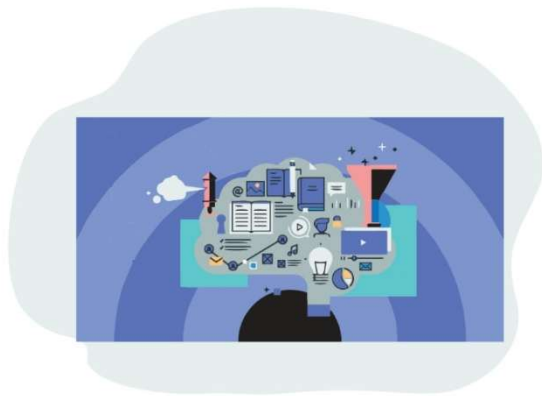
Data Science

One of the biggest reasons for Python's popularity today is its dominance in Data Science.

Python offers libraries for data processing, analysis, and visualization.

It has become the go-to language for data scientists because of its simplicity + powerful tools.

Skills in Python + Data Science are in very high demand in today's industry.



Machine Learning

Machine Learning might sound complicated, but Python makes it accessible even for beginners. With libraries like Keras and Scikit-learn, you can build powerful ML models without diving deep into complex math and algorithms.

This makes Python a top choice for anyone who wants to step into AI and ML.

Interpreted Python & Its Versions:

Of course, we know that Python is a programming language. But did you know there are two main types of programming languages?

1. Compiled Language
2. Interpreted Language

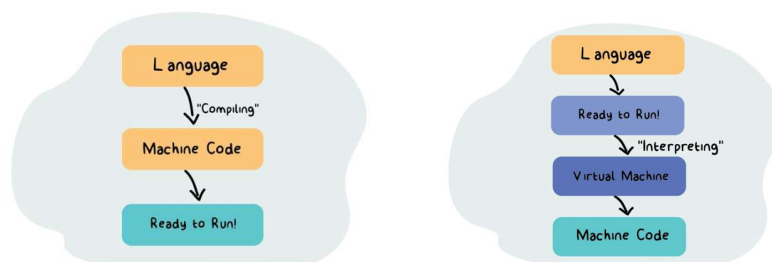
Chances of Error

- Compiled Languages:

In a compiled language, your program is first converted into machine code before it runs. If there are any errors in your code, the compiler won't let the program execute until all errors are fixed. This means you see all errors at once before running the program.

- Interpreted Languages:

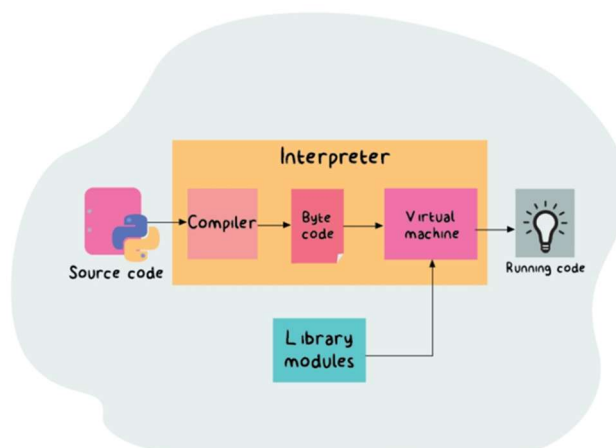
An interpreted language starts running the code immediately, even if there are errors in parts of the program. When an error is encountered, the program stops execution, and an error message shows the problem. This allows you to test code line by line.

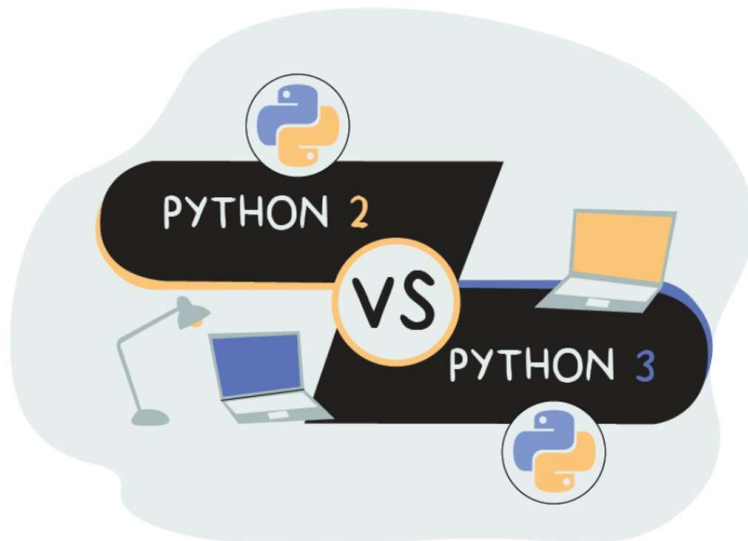


Python is an Interpreted Language

Python is a classic example of an interpreted programming language.

- Most Python implementations execute instructions directly without compiling the entire program into machine code.
- The interpreter translates each statement into subroutines and then into machine code as the program runs.
- This makes Python flexible, interactive, and beginner-friendly, because you can write and test code quickly.





Python Versions

- **Python 2 → Legacy Version**
 - Python 2 was popular for over a decade, and some companies still use it.
 - However, it is slowly becoming obsolete.
- **Python 3 → Current and Future Version**
 - Python 3 fixes many of Python 2's limitations and is actively supported.
 - For beginners, it is recommended to learn Python 3, as most modern development uses it.

Installing Python & Setting up IDEs

Installing Python:

Before you start coding, you need to have Python installed on your computer. Follow these steps:

Step 1: Check if python is installed

1. Open **Command Prompt (Windows)** or **Terminal (Mac/Linux)**.
2. Type:

```
bash
python --version
```

3. You should see the installed Python version
4. If not then follow the below steps

Step 2: Download Python

1. Go to the official website: <https://www.python.org/downloads/>
2. Click **Download Python 3.x** (latest stable version).

Step 3: Install Python

1. Run the downloaded installer.
2. **Important:** Check the box “**Add Python to PATH**” before clicking **Install Now**.
 - This makes Python accessible from the command line.
3. Follow the installation prompts.

Step 4: Verify Installation

1. Open **Command Prompt (Windows)** or **Terminal (Mac/Linux)**.
2. Type:

```
bash
python --version
```

3. You should see the installed Python version, e.g., Python 3.12.0.

Python is now ready to run programs!

Installing PyCharm IDE:

PyCharm is a popular Python IDE (Integrated Development Environment) that makes coding easier for beginners and professionals.

Step 1: Download PyCharm

1. Go to <https://www.jetbrains.com/pycharm/download/>
2. Choose Community Edition (free) and download it.

Step 2: Install PyCharm

1. Run the installer.
2. Select installation location (default is fine).
3. Check “Add launchers to PATH” if available.

4. Finish installation.

Step 3: Launch PyCharm and Configure

1. Open PyCharm.
2. On first launch, you can choose the UI theme (light or dark).
3. Create a new project:
 - Go to File → New Project
 - Choose Python Interpreter → Select the installed Python version
 - Click Create

Writing Your First Python Program in PyCharm:

1. Right-click your project folder → **New** → **Python File**
2. Name the file hello.py
3. Type the following code:

```
1 print("Hello, World!")
```

4. Right-click inside the code editor → **Run 'hello'**
5. You should see the output:

```
Hello, World!  
  
Process finished with exit code 0
```

Congratulations! You have successfully set up Python and PyCharm and run your first program.

Writing and Running Python Programs

Once Python is installed, you can start writing and running programs in multiple ways — no IDE required.

Using Python Interactive Shell (REPL):

Python comes with an **interactive mode**, also called the **REPL** (Read-Evaluate-Print Loop). It's great for experimenting and testing small pieces of code.

How to open Python REPL:

- Windows: Open Command Prompt → type `python` → press Enter
- Mac/Linux: Open Terminal → type `python3` → press Enter

You should see something like:

```
Python 3.x.x (default, ... )
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

Example:

```
>>> print("Hello, Python!")
Hello, Python!
>>> 5 + 7
12
```

The `>>>` is called the **Python prompt**, where you can type commands directly.

Using a Python Script File:

For longer programs, you usually write your code in a **Python script file** (.py) and then run it.

Steps:

1. Open any text editor (Notepad, VS Code, etc.).
2. Write your Python code, for example:

```
print("Hello, Python!")
x = 5
y = 10
print("Sum =", x + y)
```

3. Save the file with a **.py extension**, e.g., `program.py`.
4. Run the program:
 - **Windows:** Open Command Prompt → navigate to the file's folder → type `python program.py` → Enter
 - **Mac/Linux:** Open Terminal → navigate to the folder → type `python3 program.py` → Enter

You should see output like:

```
Hello, Python!
Sum = 15
```

Key Points for Beginners

- Python programs can be written and run without an IDE.
- Interactive Shell is ideal for small tests and learning concepts.
- Python script files are used for larger programs and permanent code.
- Always save your files with .py extension.

PYTHON BASICS

Python Syntax & Indentation

What is Syntax?

Every programming language has a set of rules that define how you should write code so that the computer can understand it. These rules are called the syntax of the language.

In Python, syntax is designed to be simple, clean, and readable, which is why beginners find it very easy to learn. Python code looks almost like plain English!

Example:

```
print("Hello, world!")
```

This is a valid Python statement. It prints a message to the screen. Unlike some other languages (like C, C++ or Java), Python doesn't need semicolons (;) at the end of statements — though you can use them if you want to put multiple statements on one line.

Statements and Lines:

Each line of code in Python is usually one statement.

Example:

```
x = 10
y = 20
print(x + y)
```

However, you can write multiple statements on a single line using semicolons:

```
x = 10; y = 20; print(x + y)
```

But this is not recommended for readability.

Indentation in Python:

In most programming languages, blocks of code (like inside an if, for, or function) are grouped using curly braces {}. But Python does it differently! Python uses indentation (spaces at the beginning of a line) to define code blocks.

For example:

```
if age >= 18:
    print("You are an adult.")
    print("You can vote.")
else:
    print("You are a minor.")
```

Here, the lines under the if and else statements are indented with 4 spaces, meaning they belong to those blocks. If you forget indentation or mix tabs and spaces, Python will show an 'IndentationError'.

Nested Indentation:

When you write blocks inside blocks, you increase the indentation level:

```
number = 10
if number > 0:
    print("Number is positive")
    if number % 2 == 0:
        print("It is also even")
    else:
        print("It is odd")
else:
    print("Number is negative")
```

Each inner block is indented more than the outer one.

This makes your code structure clear and readable.

Best Practices (PEP 8 Style Guide):

- Use 4 spaces per indentation level (no tabs).
- Be consistent throughout your code.
- Python will not tolerate inconsistent indentation!

Line Continuation:

Sometimes, your statement is too long to fit in one line. You can use a backslash (\) to continue it on the next line:

```
result = 100 + 200 + 300 + \
        400 + 500
```

You can also use parentheses (), brackets [], or braces { } for automatic continuation:

```
numbers = [
    1, 2, 3,
    4, 5, 6
]
```

Whitespace in Python:

Extra spaces around operators or inside parentheses don't affect execution.

```
value = ( 1 + 2 )
print(value)
```

But unnecessary spaces can make code look messy — so keep it neat!

Empty Blocks and the pass Statement

Python does not allow **empty code blocks**. If you need a placeholder for future code, use the pass statement.

Example:

```
if True:
    pass # does nothing, avoids syntax error
```


Comments in Python:

What are Comments?

Comments are notes written inside a program that are ignored by the Python interpreter. They are used to explain code, make it readable, and help others (or your future self) understand what the program does.

When Python runs your code, it skips all comments completely — they are meant only for humans, not computers.

Why Use Comments?

Comments make your code self-explanatory. They are useful for:

- Describing what a block of code does
- Explaining complex logic
- Leaving reminders or “to-do” notes
- Temporarily disabling (debugging) lines of code
- Adding author or version information

Example (conceptually):

```
# This code calculates the area of a rectangle
```

Good comments help others understand your program without needing to read every single line.

Types of Comments in Python

Python supports two types of comments:

1. Single-line comments:

A single-line comment begins with the hash symbol (#). Anything after # on that line is ignored by Python.

Example:

```
# This is a single-line comment
print("Hello, World!") # This is also a comment after a statement
```

You can place a comment on its own line or after code on the same line.

2. Multi-line comments:

Python doesn't have a special syntax for multi-line comments like some languages (e.g., /* ... */ in C). Instead, you can use triple quotes (""" or ''') to write multiple lines of text that Python will ignore if they're not assigned to a variable or used as a docstring.

Example:

```
""" -----comments----- """
''' -----comments----- '''
```

Lesson Program: Comments in python.

Source Code:

comments.py

```
1  # Python Program: Comments
2
3  # ----- SINGLE LINE COMMENTS -----
4  # This is a single-line comment
5  print("Hello, World!")  # You can also write a comment after code
6
7  # ----- MULTI-LINE COMMENTS -----
8  '''
9  This is a multi-line comment.
10 You can write as many lines as you want here.
11 Python will ignore these lines.
12 '''
13
14 """
15 Another way to write multi-line comments
16 is to use triple double quotes.
17 This is also ignored by Python.
18 """
19
20 # ----- PRACTICAL USE -----
21 # Storing user details
22 name = "John"  # student's name
23 age = 21       # student's age
24
25 print("Name:", name)
26 print("Age:", age)
```

Keywords and Identifiers:

What are Keywords?

Keywords are special reserved words in Python that have predefined meanings and purposes. They form the core vocabulary of the Python language and are used to define its syntax and structure.

For example, words like `if`, `else`, `while`, `for`, `class`, `True`, and `False` are keywords.

Each of these has a special meaning and function in the Python language.

For instance:

- `if` → used for conditional statements
- `for` → used for loops
- `def` → used to define a function
- `class` → used to define a class
- `True` and `False` → represent boolean values

Python keywords are case-sensitive, so `True` and `true` are not the same — `True` is valid, but `true` will cause an error. You can view all the keywords in Python using the built-in keyword module. This helps check which words are reserved and cannot be used as variable names.

Tip: You can check all keywords by writing `import keyword` and then `print(keyword.kwlist)` in your Python shell.

What are Identifiers?

Identifiers are names used to identify variables, functions, classes, modules, or other objects in Python.

For example:

- `age = 20` → here, `age` is an identifier.
- `def greet():` → here, `greet` is an identifier (the name of the function).

Identifiers make it easy for you (and Python) to refer to data or functions by name.

Rules for Writing Identifiers

To ensure your names are valid, Python follows these rules for identifiers:

1. Allowed Characters:
 - Can contain letters (A–Z, a–z), digits (0–9), and underscores (`_`)
 - Example: `name`, `student_1`, `_score`
2. Cannot Start with a Digit:
 - ❌ Invalid: `2name`, `123abc`

-  Valid: `name2`, `a123`
3. Cannot Use Keywords:
 - You cannot name something `if`, `class`, or `def` because they are reserved words.
 4. Case Sensitive:
 - `city` and `City` are two different identifiers.
 5. No Special Characters:
 - Characters like `@`, `#`, `-`, `$` are not allowed.
 - Example:  `my-name` is invalid.
 6. Good Practice (PEP 8):
 - Use lowercase letters for variable names (`total_marks`)
 - Use CapitalizedWords for class names (`StudentRecord`)
 - Use UPPERCASE for constants (`PI = 3.14`)

Tip: Checking Identifiers Python provides a simple method called `.isidentifier()` to check if a string is a valid identifier. Similarly, the `keyword` module helps to verify if a word is a Python keyword or not.

Lesson Program: Keywords and Identifiers in python.

Source Code:

keywordsAndIdentifiers.py

```
1  # Python Keywords and Identifiers
2
3  import keyword  # module to check Python keywords
4
5  # ----- KEYWORDS -----
6  print("=== Keywords in Python ===")
7  print("Some keywords are:", keyword.kwlist[:10])
8  print("Total number of keywords:", len(keyword.kwlist), "\n")
9
10 # ----- IDENTIFIERS -----
11 print("=== Identifiers in Python ===")
12
13 name = "Alice"
14 age1 = 21
15 _student = "John"
16 print("Valid identifiers ->", name, age1, _student)
17
18 city = "Mumbai"
19 City = "Delhi"
20 print("\ncity:", city)
21 print("City:", City)
22
23 print("\nCheck if a word is identifier or keyword:")
24 words = ["name", "2abc", "class", "value_1"]
25 for w in words:
26     print(w, "-> isidentifier?", w.isidentifier(), "| iskeyword?", keyword.iskeyword(w))
```

Variables and Constants:

Lesson Program: Variables in Python.

Source Code:

variables.py

```
1 # Python Variables
2
3 # Variables store data values
4 name = "Alice"
5 age = 21
6 height = 5.4
7
8 print("Name:", name)
9 print("Age:", age)
10 print("Height:", height)
11
12 # Variables can be reassigned
13 age = 22
14 print("\nAfter reassignment, Age:", age)
15
16 # Valid variable name example
17 _valid_name = "This is valid"
18 print("\nValid variable name example:", _valid_name)
19
20 # Variable with underscore
21 my_name = "Alice Smith"
22 print("Variable with underscore:", my_name)
23
24 # Case-sensitive variables
25 city = "Mumbai"
26 City = "Delhi"
27 print("\ncity:", city)
28 print("City:", City)
29
30 # Multiple variables declared in one line
31 x, y, z = 10, 20, 30
32 print("\nMultiple variables -> x:", x, " y:", y, " z:", z)
33
34 # Same value to multiple variables
35 a = b = c = "Python"
36 print("Same value assigned -> a:", a, " b:", b, " c:", c)
```